Explanation of the Invariant $r = \alpha x - \beta y$
and Arithmetic Progressions in the Proof of Conjecture 3

Dirk Kunert * Claude Opus 4.6†

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1 Arithmetic Progressions

Definition 1. An *arithmetic progression* is a sequence of numbers in which the difference between consecutive terms is constant. This constant is called the *common difference.*

General form. An arithmetic progression with initial value a and common difference d is the sequence

$$a,\ a+d,\ a+2d,\ a+3d,\ \ldots$$

or, as a set indexed by $k \in \mathbb{Z}$:

$$\{\, a + kd \ :\ k \in \mathbb{Z} \,\}.$$

Example 1. The sequence $3, 7, 11, 15, 19, \ldots$ is an arithmetic progression with initial value $a = 3$ and common difference $d = 4$.

Role in the proof. In the proof of Conjecture 3, each set

$$P_r \ =\ \left\{\, c_r + k(\alpha^2 + \beta^2) \ :\ k \in \mathbb{Z} \,\right\}$$

is an arithmetic progression with initial residue c_r and common difference $D = \alpha^2 + \beta^2$. The projected values $\tilde{p} = \beta x + \alpha y$ for a fixed value of r are therefore equally spaced with spacing $\alpha^2 + \beta^2$.

There are $\alpha + \beta + 1$ such progressions (one for each value of r), all sharing the same common difference but having different offsets c_r. The interleaving of these progressions — sorting all their elements together — produces the periodic difference sequence whose period length is $\alpha + \beta + 1$.

2 Why $r = \alpha x - \beta y$ Is Called an Invariant

The quantity $r := \alpha x - \beta y$ is called an *invariant* because it remains constant as one moves along a given arithmetic progression of lattice points.

Proposition 1. *Within a fixed progression P_r, the lattice points are parametrised by $k \in \mathbb{Z}$ as*

$$x \ =\ x_0 + \beta k, \qquad y \ =\ y_0 + \alpha k,$$

where (x_0, y_0) is a base point with $\alpha x_0 - \beta y_0 = r$. Then $\alpha x - \beta y = r$ for every k.

*Independent Researcher, Email: dirk.kunert@gmail.com
†AI research assistant (Anthropic).

Proof. Substituting the parametrisation:

$$\alpha(x_0 + \beta k) - \beta(y_0 + \alpha k) \;=\; \alpha x_0 - \beta y_0 + \underbrace{\alpha\beta k - \beta\alpha k}_{=0} \;=\; r.$$

The k-dependent terms cancel exactly because the step direction (β, α) is the direction of the line $f(x) = \frac{\alpha}{\beta}\,x$. $\square$

Geometric interpretation. The line $f(x) = \frac{\alpha}{\beta}\,x$ has direction vector (β, α). Moving along this direction changes x by β and y by α, leaving $\alpha x - \beta y$ unchanged. The quantity r therefore labels which of the $\alpha + \beta + 1$ parallel "tracks" a lattice point belongs to. Points on the same track share the same r and lie on the same arithmetic progression of $\tilde{p}$-values. Points on different tracks have different values of r.

Example 2. For $\alpha = 2$, $\beta = 1$, and $\omega = 1$, the invariant takes values in $\mathcal{R} = \{-1, 0, 1, 2\}$, giving $|\mathcal{R}| = 4 = 2 + 1 + 1 = \Lambda_{2,1}$.

- $r = -1$: e.g. $(x, y) = (1, 3)$, since $2\cdot 1 - 1\cdot 3 = -1$. Next point on same track: $(1+1, 3+2) = (2, 5)$, and indeed $2 \cdot 2 - 1 \cdot 5 = -1$.
- $r = 0$: e.g. $(x, y) = (1, 2)$, since $2 \cdot 1 - 1 \cdot 2 = 0$.
- $r = 1$: e.g. $(x, y) = (1, 1)$, since $2 \cdot 1 - 1 \cdot 1 = 1$.
- $r = 2$: e.g. $(x, y) = (1, 0)$, since $2 \cdot 1 - 1 \cdot 0 = 2$.

Each of these four tracks produces its own arithmetic progression of $\tilde{p}$-values with common difference $D = 4 + 1 = 5$. Their interleaving gives the difference sequence $(1, 1, 1, 2)$ with period $4 = \Lambda_{2,1}$.